

# B A 286T Statistics (TEMBA)

Unique numbers 02885 (Cohort 1), 02890 (Cohort 2) — Fall 2026

Jared S. Murray

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**Office hours:** TBD (announced on Canvas in the first week), and by appointment

**Class meetings:** Cohort 1 meets Mondays in RRH 4.408; Cohort 2 meets Tuesdays in RRH 4.416. Dates are in the Course Outline below.

**Course website:** Canvas ([utexas.instructure.com](https://utexas.instructure.com)).

## Course Description

**University catalog description:** A unified approach to basic concepts in collection, analysis, and interpretation of data, emphasizing capabilities of different statistical methods and business applications. Students use statistical software packages.

### What will I learn?

On successfully completing this course you will be able to

1. Summarize and visualize data to describe variables and the relationships between them
2. Use the language of probability to understand and communicate risk, and to make decisions under uncertainty
3. Use data to draw appropriate inferences, test hypotheses, and estimate the parameters of probability models used for prediction, understanding, and decision making
4. Report the results of a data analysis with appropriate measures of uncertainty and error
5. Conduct regression analyses for prediction and for understanding relationships between predictors and an outcome

### Leadership and this course

The Texas McCombs Hildebrand MBA program is designed to develop influential business leaders. The MBA Program has identified four fundamental and broad pillars of leadership: knowledge and understanding, communication and collaboration, responsibility and integrity, and a worldview of business and society.

In this course, you will learn how to extract actionable insights from data, and to use them to make predictions and decisions under uncertainty. You will learn how to communicate the results of your analyses and how to critically evaluate the results of data analyses presented to you.

### How will I learn?

This course is fully synchronous and in-person. I record lectures, but this isn't a hybrid class and there is no remote participation except in extenuating circumstances.

## Communication

Your student email address is EID-based: [your eid]@my.utexas.edu, and will be listed as your primary email address in your student record. **All important University communications will be sent to this email address.**

The course Canvas site can be found at [utexas.instructure.com](https://utexas.instructure.com). The fastest way to reach me is email, at [jared.murray@mcombs.utexas.edu](mailto:jared.murray@mcombs.utexas.edu). Canvas sends announcements and notifications to your school-issued email, so check it regularly.

The University of Texas at Austin expects all students to check their official email regularly to stay current with university communications. Failure to check your email is not an acceptable excuse for missing important information.

## Asking for help

I encourage you to work together to help each other understand the course material, and you may work together on homework assignments. You may also get help by attending my office hours or scheduling an appointment with me.

## Prerequisites

Graduate standing and admission to the McCombs School of Business. (Restricted to Texas Evening MBA.)

## How to succeed in this course

The most important thing you can do is be present, engaged, and ask lots of questions. We will move quickly through some challenging technical material. Start working on practice problems and homework assignments soon after class, while the material is still fresh and you have time to get help. If you feel yourself getting behind, come talk to me and we'll make a plan to get back on track.

**Statement on Academic Rigor.** At McCombs, *you can expect to feel intellectually challenged in your classes*. This may be experienced as ambition, boldness, curiosity, doubt, excitement, fear, gratitude, or something else entirely. We encourage you to embrace these feelings as indicators of authentic learning and growth. Our mission is to nurture the intellectual exploration and academic tenacity that will help you build the knowledge, skills, and values as foundations to thrive in your forthcoming professional pursuits.

To achieve this mission, we implement a rigorous curriculum designed to guide you in mastering complex concepts across diverse disciplines and contexts. We emphasize high standards and empower you to develop the in-demand skills needed for success in today's workforce as well as the durable skills required to adapt and lead tomorrow's workforce.

Although it is normal to experience difficult coursework in your McCombs classes, we are dedicated to supporting your development and academic success. Our goal is to help each of you live up to your potential and leave here knowing what it means to produce high-quality, credible work. Our priority is that each of you who starts here at McCombs grows to become a curious, life-long learner equipped to change the world.

## Course Requirements

### Required materials

There is no textbook to buy. Course notes, slides, data, and assignments are posted on Canvas and the course website.

### Software and computing

We will use AI tools to drive statistical computing programming languages (mostly R, maybe Python) for our work in class and on homework assignments. UT gives you institutional access to ChatGPT and Claude with

plenty of usage to complete our work and still utilize them for other things. This is a pilot; these institutional accounts are new. There might be some changes in how we use them as the course progresses. The important thing to know is that they are tools for producing analytic *inputs*: data summaries, visualizations, model fits, and so on. We will not use them to choose methods, plan analyses, or to interpret results.

We are using these tools to remove one of the biggest pain points in a class like this: industrial-strength statistical analysis doesn't happen in Excel or via expensive software with limited functionality, and this is not the place to teach statistical programming. However, if you want to learn some statistical programming our AI workflow is great for that too. And regardless, you will learn some generalizable skills for how to do quantitative work with AI.

Your ability to complete ANY assignment, quiz, or exam will never depend on a functioning AI tool being available to you. I will provide known good tables and figures that you can use if the AI tools aren't working properly, or if you just want to check your work. You will be evaluated on concepts, not code.

There is nothing to install before the first class: we will set up the tools together during the August residency. Just bring a laptop.

### **Required devices**

Bring a laptop to class.

### **Confidentiality of class recordings**

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

### **Getting help with technology**

Students needing help with technology in this course should contact the [UT Service Desk](#).

### **Classroom expectations**

**Class attendance.** I expect everyone to be in class, unless there are extenuating circumstances. I won't take attendance, but you will struggle if you aren't regularly attending.

**Class participation.** I expect you to be active and engaged, and ideally asking questions; we will spend time in discussion and working through problems together or in small groups, and you will get the most out of this experience by participating fully.

**Behavior expectations.** I expect everyone to conduct themselves professionally and to treat one another with respect, and I will do the same with you.

**McCombs Classroom Professionalism Policy.** The highest professional standards are expected of all members of the McCombs community. The collective class reputation and the value of the Texas MBA experience hinges on this. You should treat the Texas MBA classroom as you would a corporate boardroom. Faculty are expected to be professional and prepared to deliver value for each and every class session. Students are expected to be professional in all respects. The Texas MBA classroom experience is enhanced when:

- Students arrive on time. On time arrival ensures that classes are able to start and finish at the scheduled time. On time arrival shows respect for both fellow students and faculty and it enhances learning by reducing avoidable distractions.
- Students display their name cards. This permits fellow students and faculty to learn names, enhancing opportunities for community building and evaluation of in-class contributions.
- Students are fully prepared for each class. Much of the learning in the Texas MBA program takes place during classroom discussions. When students are not prepared, they cannot contribute to the overall learning process. This affects not only the individual, but their peers who count on them, as well.

- Students respect the views and opinions of their colleagues. Disagreement and debate are encouraged. Intolerance for the views of others is unacceptable.
- Students do not confuse the classroom for the cafeteria. The classroom (boardroom) is not the place to eat your breakfast tacos, wraps, sweet potato fries, or otherwise set up for a picnic. Please plan accordingly. Recognizing that back-to-back classes sometimes take place over the lunch hour, energy bars and similar snacks are permitted. Please be respectful of your fellow students and faculty in your choices.
- Students minimize unscheduled personal breaks. The learning environment improves when disruptions are limited.
- Students attend the class section to which they are registered. Learning is enhanced when class sizes are optimized. Limits are set to ensure a quality experience and safety.
- Technology is used to enhance the class experience. When students are surfing the web, responding to e-mail, instant messaging each other, and otherwise not devoting their full attention to the topic at hand they are doing themselves and their peers a major disservice. Those around them face additional distraction. Fellow students cannot benefit from the insights of the students who are not engaged. Faculty office hours are spent going over class material with students who chose not to pay attention, rather than truly adding value by helping students who want a better understanding of the material or want to explore the issues in more depth. Students with real needs may not be able to obtain adequate help if faculty time is spent repeating what was said in class. There are often cases where learning is enhanced by the use of technology in class. Faculty will let you know when it is appropriate.
- Phones and wireless devices are turned off. We’ve all heard the annoying ringing in the middle of a meeting. Not only is it not professional, it cuts off the flow of discussion when the search for the offender begins. When a true need to communicate with someone outside of class exists (e.g., for some medical need) please inform the professor prior to class.

## Artificial intelligence

You will use generative AI tools (ChatGPT, Claude, Copilot, and the like) on homework assignments, within strict bounds outlined in those assignments. They are primarily to be used as natural language interfaces to statistical programming languages. You shouldn’t use them blindly to interpret analyses. You may use them for limited other purposes like formatting your assignments or proofreading text you wrote, and using them as “study companions” is also acceptable. As a rule of thumb: If it would feel dishonest to ask another person to do what you want the AI to do, then it’s probably cheating.

1. **You are responsible for everything you turn in.** If you use AI to help write R code or prose, you must be able to explain what it does and why it answers the question. Submitting homework you don’t understand is the fastest way to do badly on the exam, which is closed to AI.
2. **Say when you used it.** Add a short note to any assignment where AI contributed substantively (in ways I didn’t design), describing what you used it for. Using generative AI without authorization, or failing to disclose its use where required, may constitute a violation of UT Austin’s Institutional Rules on academic integrity and may be referred to the Office of Student Conduct and Academic Integrity.

Generative AI is **not** permitted on knowledge checks or the final exam. If you are unsure whether a use is allowed, ask me first.

## Assignments and grading

| Assignment              | Percent of total grade |
|-------------------------|------------------------|
| Homework assignments    | 40                     |
| Knowledge check quizzes | 25                     |
| Final exam              | 35                     |

## Course outline

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at [utexas.instructure.com](https://utexas.instructure.com). Check this site regularly.

**Changes** to the schedule may be made at my discretion and if circumstances require. It is your responsibility to note these changes when announced (although I will do my best to ensure that you receive the changes with as much advance notice as possible).

| Date (C1 / C2) | Class topic                                                             | Assessments due   |
|----------------|-------------------------------------------------------------------------|-------------------|
| 8/23           | Introduction; describing and visualizing data                           |                   |
| 8/24           | Probability basics; distributions, expectation and variance; the normal |                   |
| 8/31 / 9/1     | Multivariate probability and portfolios                                 | Homework 1        |
| 9/9 / 9/8      | Statistical inference and hypothesis testing                            | Knowledge check 1 |
| 9/14 / 9/15    | Simple linear regression                                                | Homework 2        |
| 9/21 / 9/22    | Multiple linear regression I                                            | Knowledge check 2 |
| 9/28 / 9/29    | Multiple linear regression II                                           | Homework 3        |
| 10/5 / 10/6    | Final exam                                                              |                   |

Note that Cohort 1 meets Wednesday 9/9 (not Monday 9/7, Labor Day).

## Policies

### Classroom policies

#### Statement on learning success

Your success in this class is important to me. We will all need accommodations because we all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course. I also encourage you to reach out to the student resources available through UT. Many are listed on this syllabus, but I am happy to connect you with a person or Center if you would like.

### Grading policies

I will assign letter grades based on the following cutoffs. While there is no pre-determined grade distribution for this class I may (and often do) curve final grades before assigning letter grades. Any curve will **only** be in your favor, and will not change your overall class rank.

| Grade | Cutoff |
|-------|--------|
| A     | 94%    |
| A-    | 90%    |
| B+    | 87%    |
| B     | 84%    |
| B-    | 80%    |
| C+    | 77%    |
| C     | 74%    |
| C-    | 70%    |
| D     | 65%    |
| F     | <65%   |

## **Late work**

Late work is only accepted under extenuating circumstances. If you're having trouble, or anticipate a problem with a deadline, reach out to me as soon as you can.

## **Absences**

There is no formal attendance policy and you do not need to inform me of any absences; however, if you anticipate missing significant amounts of class due to illness or emergency please let me know as soon as possible.

## **Religious holy days**

By [UT Austin policy](#), you must notify me of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence. For questions regarding religious accommodations, please contact the [Office of the Dean of Students](#).

## **Sharing of course materials is prohibited**

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

## **Student rights and responsibilities**

- You have a right to a learning environment that supports mental and physical wellness.
- You have a right to respect.
- You have a right to be assessed and graded fairly.
- You have a right to freedom of opinion and expression.
- You have a right to privacy and confidentiality.
- You have a right to meaningful and equal participation, to self-organize groups to improve your learning environment.
- You have a right to learn in an environment that is welcoming to all people. No student shall be isolated, excluded or diminished in any way.

With these rights come responsibilities:

- You are responsible for taking care of yourself, managing your time, and communicating with the teaching team and with others if things start to feel out of control or overwhelming.
- You are responsible for acting in a way that is worthy of respect and always respectful of others. Your experience with this course is directly related to the quality of the energy that you bring to it, and your energy shapes the quality of your peers' experiences.
- You are responsible for creating an inclusive environment and for speaking up when someone is excluded.
- You are responsible for holding yourself accountable to these standards, holding each other to these standards, and holding the teaching team accountable as well.

## University policies

### Academic integrity

Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, suspension, or expulsion from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at <http://deanofstudents.utexas.edu/conduct>.

### University resources for students

Your success in this class is important to me. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. There are also a range of resources on campus:

**Disability and access.** This class respects and welcomes students of all backgrounds, identities, and abilities. If there are circumstances that make our learning environment and activities difficult, if you have medical information that you need to share with me, or if you need specific arrangements in case the building needs to be evacuated, please let me know. I am committed to creating an effective learning environment for all students, but I can only do so if you discuss your needs with me as early as possible. I promise to maintain the confidentiality of these discussions. If appropriate, also contact Disability and Access at 512-471-6259, <https://disability.utexas.edu/about/>.

**Counseling and Mental Health Center.** Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. <https://healthyhorns.utexas.edu/cmhc/>

**Student Outreach and Support.** If at any time you experience an emergency that necessitates your absence from a class requirement (e.g., attendance, assignment submission, or exam), please report your circumstances and absence via the Student Outreach and Support website: <https://deanofstudents.utexas.edu/sos/index.php>

### Title IX reporting

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university's [relevant policies](#).

Beginning January 1, 2020, Texas [Senate Bill 212 \(SB 212\)](#) requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents** that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a

responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email [advocate@austin.utexas.edu](mailto:advocate@austin.utexas.edu). For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at [titleix@austin.utexas.edu](mailto:titleix@austin.utexas.edu), or call 512-471-0419.

## Important safety information

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right — it probably isn't. Trust your instincts and share your concerns.

The following recommendations regarding emergency evacuation from the Security and Emergency Management Office, <https://www.utexas.edu/campus-life/safety-and-security>:

Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.

- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- Information regarding emergency evacuation routes and emergency procedures can be found at [emergency.utexas.edu](https://www.utexas.edu/emergency).

## Carrying of handguns on campus

Students in this class should be aware of the following university policies related to Texas' Open Carry Law:

- Students in this class who hold a license to carry are asked to [review the university policy regarding campus carry](#).
- Individuals who hold a license to carry are eligible to carry a concealed handgun on campus, including in most outdoor areas, buildings and spaces that are accessible to the public, and in classrooms.
- It is the responsibility of concealed-carry license holders to carry their handguns on or about their person at all times while on campus. Open carry is NOT permitted, meaning that a license holder may not carry a partially or wholly visible handgun on campus premises or on any university driveway, street, sidewalk or walkway, parking lot, parking garage, or other parking area.